

Modular arithmetic

Outline

Euclid's division

Modulo and modulo congruence

Euclid's division

Introduction: Euclid's division

- ▶ In this section we recall the **division with remainder** between integer numbers
- ▶ We prove that quotient and remainder always exist and are unique
- ▶ From it we define **multiples**, **factors** and **divisibility**, the basic notions of modular arithmetic

Lemma 1: Euclid's division (or division with remainder)

For every pair of integer numbers n and m , with $m \neq 0$, there exist unique integer numbers q and r such that

$$n = q m + r \quad \text{and} \quad 0 \leq r < |m| \quad (1)$$

- ▶ The value n is called the **dividend**, the value m is called the **divisor**, the value q is called the **quotient** and the value r is called the **remainder**.
- ▶ The standard convention is that the remainder is non negative but there are other conventions where the remainder r can be negative.

Example

- ▶ For $n = 17$ and $m = 5$ we have:

$$17 = 3 \cdot 5 + 2, \quad q = 3, \quad r = 2$$

- ▶ For $n = -17$ and $m = 5$ we have:

$$-17 = (-4) \cdot 5 + 3, \quad q = -4, \quad r = 3$$

(and not $q = -3$, $r = -2$, since the remainder must satisfy $0 \leq r < |m|$)

Proof (1/3)

To prove *existence*, we consider the set

$$S = \{ n - km : k \in \mathbb{Z} \text{ and } n - km \geq 0 \}$$

- ▶ The set S is nonempty, since for $k = -|n| m$ we have

$$n - km = n + |n| m^2 \geq n + |n| \geq 0$$

- ▶ Since S is a nonempty subset of the nonnegative integers, the well-ordering principle guarantees that there exists a smallest element $r \in S$, meaning there exists some integer q such that

$$r = n - qm, \quad \text{with } r \geq 0$$

Proof (2/3)

- By construction, we must have $r < |m|$, since if $r \geq |m|$, then we could write

$$r' = r - |m| = n - \left(q + \frac{|m|}{m}\right) m \geq 0$$

which would imply $r' \in S$, contradicting the minimality of r .

Proof (3/3)

To prove *uniqueness*, suppose there exist two pairs (q_1, r_1) and (q_2, r_2) such that

$$n = q_1m + r_1 = q_2m + r_2$$

with $0 \leq r_1, r_2 < |m|$.

- ▶ Subtracting the two equations, we have

$$(q_1 - q_2)m = r_2 - r_1$$

- ▶ Since $|r_2 - r_1| < |m|$ and m divides $r_2 - r_1$ (which equals the left-hand side), the only possibility is $r_1 = r_2$, which implies $q_1 = q_2$, ensuring uniqueness.



Definition 1: multiple

*An integer number n is a **multiple** of an integer number m if there exists an integer number q such that*

$$n = q \cdot m \quad (2)$$

Definition 2: factor

*An integer number m is a **factor** of an integer number n if there exists an integer number q such that*

$$n = q \cdot m \quad (3)$$

- ▶ An integer number n is **divisible** by an integer number m if m is a factor of n .

Summary: Euclid's division

Division with remainder:

- ▶ For $n, m \in \mathbb{Z}$, $m \neq 0$: $n = qm + r$ with $0 \leq r < |m|$
- ▶ Quotient q and remainder r exist and are unique

Multiples, factors, divisibility:

- ▶ n is a multiple of m (and m is a factor of n) if $n = q \cdot m$ for some $q \in \mathbb{Z}$
- ▶ n is divisible by m if m is a factor of n , i.e., the remainder is 0

Modulo and modulo congruence



Introduction: Modulo and modulo congruence

- ▶ We now give a name to the remainder of Euclid's division: the **modulo** operation
- ▶ Two numbers with the same remainder are called **congruent**
- ▶ Modular arithmetic is used, for instance, in cyclic phenomena (clocks, calendars) and in computer science (hashing, cryptography)

Modulo

Given two integer numbers n and m , with $m \neq 0$, n **modulo** m (abbreviated in $n \bmod m$) is the **remainder** r of the Euclidean division (or division with remainder) of n by m , where n is the **dividend** and m is the **divisor**.

- It follows from the definition that:

$$0 \leq n \bmod m < |m| \quad \forall n, m \in \mathbb{Z}, m \neq 0$$

Congruence modulo m

Two natural numbers n and k are **congruent** modulo a natural number $m \geq 1$ if

$$n \bmod m = k \bmod m$$

and we write

$$n \equiv k \pmod{m}$$

- ▶ The parentheses mean that $(\bmod m)$ applies to the entire equation, not just to the right-hand side (here, k).

Equivalent characterizations

Equivalently, we have $n \equiv k \pmod{m}$

1. if n and k have the same remainder when divided by m .
2. if m is a divisor of $|n - k|$, i.e., if there exists a natural number q such that $|n - k| = q \cdot m$ or equivalently if there exists an integer number q such that $n - k = q \cdot m$

Example

- ▶ Congruence modulo 5: 23 and 13 are congruent modulo 5 and we have

$$23 \equiv 13 \pmod{5}$$

- ▶ Indeed, $23 \bmod 5 = 3 = 13 \bmod 5$, and $23 - 13 = 10 = 2 \cdot 5$

Summary: Modulo and modulo congruence

Modulo:

- ▶ $n \bmod m$ is the remainder of the division of n by m
- ▶ $0 \leq n \bmod m < |m|$

Congruence:

- ▶ $n \equiv k \pmod{m}$ if $n \bmod m = k \bmod m$
- ▶ Equivalently, m divides $n - k$

Key Takeaways

Summary

1. Euclid's division:

- ▶ $n = qm + r$ with $0 \leq r < |m|$, with q and r unique
- ▶ Existence via the well-ordering principle, uniqueness via $|r_2 - r_1| < |m|$

2. Divisibility:

- ▶ Multiples and factors: $n = q \cdot m$ with $q \in \mathbb{Z}$

3. Modular arithmetic:

- ▶ $n \bmod m$ is the remainder; $n \equiv k \pmod{m}$ means same remainder